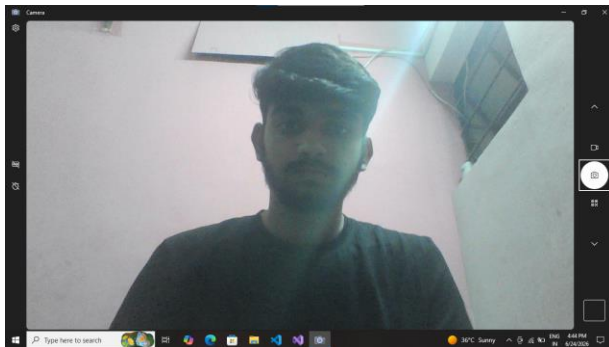


Day 4 :- Connecting and Testing Peripherals Practical : Set up and test peripherals like printers and webcams . Task : Resolve connectivity issues for a give setup.

Tasks 1 :- Connect a USB webcam to your computer, install any required drivers, and verify that the webcam works by opening the default camera app and capturing a photo.

Ans. :-



➤ **USB Webcam Installation and Verification Steps :-**

1. Connect the Webcam :-

- Plug the USB webcam into an available USB port on your computer.
- Wait for Windows to detect the device.

2. Install Drivers :-

- Most webcams install automatically through Windows.
- If prompted, follow the on-screen instructions.
- If drivers are not found, download them from the webcam manufacturer's website.

3. Verify Device Detection :-

- Press Windows + X → Device Manager.
- Expand Cameras or Imaging Devices.
- Confirm that the webcam appears without any warning symbols.

4. Open the Camera App :-

- Click Start and search for Camera.
- Open the Camera application.

5. Test the Webcam :-

- Check that a live video preview appears.
- Adjust the webcam position if needed.

6. Capture a Photo :-

- Click the Photo button (camera icon).
- The image will be saved automatically in the Pictures > Camera Roll folder.

Tasks 2 :- Set up a local printer (USB or network), install the correct driver, and print a test page to confirm successful installation.

Ans. :-

➤ Printer Setup and Test Page Printing :- 1. Connect the Printer :-

USB Printer

- Connect the printer to the computer using a USB cable.
- Turn on the printer.

Network Printer

- Connect the printer to the same Wi-Fi network as the computer.
- Ensure the printer is powered on.

2. Install the Printer Driver :-

- Open Settings → Bluetooth & devices → Printers & scanners.
- Click Add device.
- Select your printer from the list.
- If Windows cannot find the driver, download the correct driver from the printer manufacturer's website and install it.

3. Verify Installation :-

- Open Device Manager or Printers & scanners.
- Confirm that the printer appears without any errors.

4. Print a Test Page :-

- Go to Settings → Bluetooth & devices → Printers & scanners.
- Select your printer.
- Click Printer properties.
- Click Print Test Page.

5. Confirm Success : -

Check that :-

- The printer receives the print job.
- A test page is printed correctly.
- There are no error messages.

**Tasks 3 :- Simulate a connectivity issue by unplugging the printer's USB cable, then use Windows Device Manager to identify and resolve the 'printer not detected' problem.

Hint: Check for error icons and use the 'Scan for hardware changes' option after reconnecting. Ans. :- ○ Simulating a "Printer Not Detected" Issue :- 1. Disconnect the Printer :-**

- Turn on the printer.
- Unplug the USB cable connecting the printer to the computer.

2. Check Device Manager :-

- Press Windows + X and select Device Manager.
- Expand Printers, Print Queues, or Universal Serial Bus Controllers.
- Look for:
 - A missing printer entry.
 - A device with a yellow warning icon (!).
 - An "Unknown Device" entry.

3. Reconnect the Printer :-

- Plug the USB cable back into the computer.
- Wait a few seconds for Windows to detect the hardware.

4. Scan for Hardware Changes :-

- In Device Manager, click Action → Scan for hardware changes.
- Windows should search for and reinstall the printer if detected.

5. Verify Detection :-

- Check that the printer appears normally in Device Manager.
- Open Settings → Bluetooth & devices → Printers & scanners.
- Confirm that the printer is listed and shows Ready status.

6. Test the Printer :-

- Open Printer Properties.
- Click Print Test Page to verify functionality.

**Tasks 4 :- A user reports that their webcam video is not displaying in Zoom, but it works in the default camera app. Troubleshoot and resolve the issue so the webcam works in both applications.

Hint: Check if another app is using the webcam or if Zoom permissions are enabled.**

Ans. :-

Troubleshooting Webcam Not Working in Zoom :-

Steps to Resolve :-

1. Close Other Applications :-

- Exit the Camera app and any other apps that may be using the webcam (Teams, Skype, Google Meet, etc.).
- Restart Zoom and test the camera.

2. Check Zoom Camera Settings :-

- Open Zoom.
- Go to Settings → Video.
- Select the correct webcam from the Camera drop-down menu.
- Verify that a video preview appears.

3. Check Windows Camera Permissions :-

- Open Settings → Privacy & Security → Camera.
- Ensure Camera access is turned On.
- Ensure Let desktop apps access your camera is On.
- Confirm that Zoom is allowed to use the camera.

4. Restart the Webcam :-

- Disconnect and reconnect the webcam (if external).
- Restart Zoom.

5. Check Device Manager :-

- Press Windows + X → Device Manager.
- Expand Cameras.
- Ensure the webcam appears without any warning icons.
- If necessary, right-click the webcam and select Update Driver.

6. Restart the Computer :-

- Reboot the system to release any application that may still be using the webcam.

